

Hacker rank Numpy

1. The *NumPy* (Numeric Python) package helps us manipulate large arrays and matrices of numeric data.

To use the *NumPy* module, we need to import it using:

```
import numpy
```

Arrays

A *NumPy* array is a grid of values. They are similar to lists, except that every element of an array must be the same type.

```
import numpy
```

```
a = numpy.array([1,2,3,4,5])
```

```
print a[1]    #2
```

```
b = numpy.array([1,2,3,4,5],float)
```

```
print b[1]    #2.0
```

In the above example, `numpy.array()` is used to convert a list into a *NumPy* array. The second argument (`float`) can be used to set the type of array elements.

Task

You are given a space separated list of numbers.

Your task is to print a reversed *NumPy* array with the element type float.

Input Format

A single line of input containing space separated numbers.

Output Format

Print the reverse *NumPy* array with type float.

Sample Input

```
1 2 3 4 -8 -10
```

Sample Output

[-10. -8. 4. 3. 2. 1.]

Change Theme

Language

Pypy 3

```
1 import numpy
2
3 def arrays(arr):
4     # complete this function
5     # use numpy.array
6     a =numpy.array(arr,float)
7     return a[::-1]
8
9 arr = input().strip().split(' ')
10 result = arrays(arr)
11 print(result)
```



You have earned 20.00 points!
You are now 45 points away from the 4th star for your python badge.

59%

175/220

Congratulations

You solved this challenge. Would you like to challenge your friends?



Next Challenge

 Test case 0

Compiler Message

 Test case 1

Success

Input (stdin)

1

1 2 3 4 -8 -10

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Expected Output

1

[-10. -8. 4. 3. 2. 1.]

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2. shape

The *shape* tool gives a tuple of array dimensions and can be used to change the dimensions of an array.

(a). Using *shape* to get array dimensions

```
import numpy
```

```
my__1D_array = numpy.array([1, 2, 3, 4, 5])  
print my_1D_array.shape  #(5,) -> 1 row and 5 columns
```

```
my__2D_array = numpy.array([[1, 2],[3, 4],[6,5]])  
print my_2D_array.shape  #(3, 2) -> 3 rows and 2 columns
```

(b). Using *shape* to change array dimensions

```
import numpy
```

```
change_array = numpy.array([1,2,3,4,5,6])  
change_array.shape = (3, 2)  
print change_array
```

#Output

```
[[1 2]
```

```
[3 4]
```

```
[5 6]]
```

reshape

The *reshape* tool gives a new shape to an array without changing its data. It creates a new array and does not modify the original array itself.

```
import numpy
```

```
my_array = numpy.array([1,2,3,4,5,6])  
print numpy.reshape(my_array,(3,2))
```

#Output

```
[[1 2]
```

```
[3 4]
```

[5 6]]

Task

You are given a space separated list of nine integers. Your task is to convert this list into a 3X3 *NumPy* array.

Input Format

A single line of input containing space separated integers.

Output Format

Print the 3X3 *NumPy* array.

Sample Input

1 2 3 4 5 6 7 8 9

Sample Output

[[1 2 3]

[4 5 6]

[7 8 9]]

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```
1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 import numpy as np
3 def arrays(arr):
4     arr1=np.array(arr,int)
5     return arr1.reshape(3,3)
6
7 arr =input().strip().split(' ')
8 res = arrays(arr)
9 print(res)
```



You have earned 20.00 points!

You are now 25 points away from the 4th star for your python badge.

77%

195/220

Congratulations

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Next Challenge

Test case 0

Compiler Message

Success

Test case 1



Test case 2



Input (stdin)

Download

```
1 1 2 3 4 5 6 7 8 9
```

Expected Output

Download

```
1 [[1 2 3]
2  [4 5 6]
3  [7 8 9]]
```